ABDUL MUQTADIR

Control Systems · Artificial Intelligence · Embedded Systems & Robotics Engineer

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RESEARCH & ENGINEERING INTERESTS

Closed-Loop Digital Control Systems (PID, State-Space, Linear Quadratic Regulator - LQR), Artificial Intelligence & Applied Machine Learning, Cyber-Physical & Embedded Systems, Sensor Fusion & Autonomous Robotics, Hardware-in-the-Loop (HIL) Flight Telemetry, and Applied RF & Microwave Engineering.

EDUCATION

National University of Sciences and Technology (NUST) Sep 2023 – Expected Jun 2027

Bachelor of Engineering in Avionics Engineering; CGPA: 3.74 / 4.00 (Semester 7) *College of Aeronautical Engineering, Risalpur*

- **Academic Honors:** Dean's Honor List (Awarded across 5 semesters).
- **Core Coursework:** Control Systems Engineering, Digital Signal Processing (DSP), Microcontrollers & Embedded Systems, Artificial Intelligence, Sensor Systems, System Dynamics & Modeling, and RF & Wave Propagation.

The Universal Millennium College Aug 2021 – May 2023

Cambridge International A-Levels — Mathematics (A), Physics (A), Chemistry (A) *Pakistan*

Bloomfield Hall School 2008 – 2021

Cambridge International O-Levels — 1 A, 6 A Grades (Science & Mathematics Stream)* *Pakistan*

TECHNICAL EXPERTISE & LABORATORY TOOLS

Control Systems & Dynamics: Closed-Loop Digital PID Control, Linear Quadratic Regulator (LQR), State-Space Modeling, IMU 6-DOF Sensor Fusion (Kalman & Complementary Filters), Lagrangian Equations of Motion, Dynamic Plant Identification, MATLAB / Simulink.

AI, Data & Programming: Python (NumPy, SciPy, Pandas, Scikit-Learn, Matplotlib), Applied Machine Learning, Predictive Modeling, C, C++, Embedded C, MATLAB, G-Code, Linux/Bash, Git/GitHub.

Embedded Systems & Firmware: ESP32 (Dual-Core 240MHz), STM32, Arduino Uno, FreeRTOS / Timer Interrupts, Low-Level Peripheral Control, Protocol Interfacing (I2C, SPI, UART, PWM).

Robotics, Mechatronics & Prototyping: MPU6050 6-DOF IMU, BTS7960 43A High-Current Motor Drivers, A4988 Stepper Drivers, NEMA Motors, Digital Storage Oscilloscopes (DSOs), CATIA V5 (3D Solid Modeling), AutoCAD, KiCAD/EasyEDA (PCB Layout).

RF, Microwave & Telemetry: CST Studio Suite 2025 (Time & Frequency Domain 3D Solvers), Keysight ADS, S-Parameters (S11 return loss, S21 isolation), Super Wideband Vivaldi Antennas, ADALM-Pluto SDR, 915 MHz FM Telemetry.

RESEARCH & ENGINEERING PROJECTS

Two-Wheeled Inverted Pendulum Robot: LQR vs. Cascaded PID Analysis May 2026

Control Systems Lead | MATLAB/Simulink, Embedded C++, MPU6050 6-DOF IMU *NUST CAE*

- Derived complete non-linear Lagrangian equations of motion and linearized state-space matrices around the unstable vertical equilibrium.
- Demonstrated physical degradation of theoretical state-feedback LQR under unobserved linear cart states (position and velocity); engineered and stabilized a physical prototype via cascaded PID firmware with anti-windup clamping and complementary sensor filtering.

Hardware-in-the-Loop (HIL) UAV Altitude Control & SDR Telemetry Jul 2026

Systems & Communications Engineer | ADALM-Pluto SDR, Python, PID Autopilot *NUST CAE*

- Constructed a real-time HIL telemetry framework coupling a simulated second-order aircraft climb dynamics plant with a closed-loop digital PID autopilot.
- Modulated continuous altitude telemetry over a physical 915 MHz RF link using an ADALM-Pluto SDR with frequency modulation (FM); built receiver DSP pipeline with quadrature demodulation and single-pole IIR noise filtering to reconstruct the autopilot feedback loop under noise.

Autonomous Heavy Competition Rover Apr 2025

Hardware & Power Engineer | ESP32 Dual-Core, Dual BTS7960 43A, 4S2P Li-Ion *Robotics R&D*

- Engineered heavy-duty powertrain utilizing dual BTS7960 43A H-bridges, JGA-25-370 geared motors, and custom 16.8V battery management with real-time wireless telemetry.

CNC 2D Handwriting Machine & Vector Plotter Dec 2025

Mechatronics Designer | Arduino Uno, GRBL v1.1, A4988, NEMA 14, CATIA V5 *Personal Project*

- Designed dual-axis mechanical gantry in CATIA V5 and calibrated GRBL firmware EEPROM to achieve 0.05 mm positional repeatability for vector G-code execution.

Discrete Multi-Waveform Analog Function Generator Mar 2026

Analog Circuit Designer | NE555 Core, Active Op-Amp Integrator & Shaper, Multisim *NUST CAE*

- Designed and physically validated a discrete multi-waveform analog generator providing square, triangle, and sine waveforms from 10 Hz to 50 kHz with total harmonic distortion (THD < 2.5%).

Super Wideband (3–18 GHz) Conformal Vivaldi Antenna for Electronic Warfare Aug 2026

Lead Designer | CST Studio Suite 2025, Rogers RT6035HTC, CPW Transition *NUST CAE*

- Designed and numerically simulated an ultra-compact planar and aerodynamically conformal Vivaldi antenna achieving a 6:1 impedance bandwidth ratio with return loss S11 < -10 dB across 3.0 to 18.0 GHz for airborne Electronic Support Measures (ESM).
- Synthesized a single-layer Co-Planar Waveguide (CPW) feed transition and exponential flare profile (opening rate R = 0.16 mm^{-1}) with 4 curved ground slits; modeled aerodynamic conformal cylindrical curvature matching UAV fuselage contours (isolation S21 < -25 dB).

2.4 GHz 5th-Order Chebyshev Stepped-Impedance Low-Pass Filter Jun 2026

RF Designer | Keysight ADS, Keysight LineCalc, CST 3D EM, FR4 Microstrip *NUST CAE*

- Synthesized a distributed stepped-impedance microstrip low-pass filter at a 2.4 GHz cutoff frequency for avionics ISM-band harmonic suppression.
- Executed Monte Carlo statistical yield analysis in Keysight ADS to model substrate permittivity variations (dielectric constant Er = 4.4 ± 0.1) and trace width tolerances; verified full 3D S-parameters in CST Studio Suite.

PROFESSIONAL EXPERIENCE & LEADERSHIP

Freelance Engineering Consultant Jul 2025 – Present

Independent Contractor — Embedded Systems, Controls & Firmware *Remote*

- Consult on microcontroller firmware architecture, I2C/SPI sensor interfaces, and closed-loop control algorithms for international engineering clients.
- Perform schematic reviews, circuit debugging, and component selection for high-reliability embedded hardware builds.

Head of Avionics Club Sep 2024 – Present

College of Aeronautical Engineering, NUST *Risalpur, Pakistan*

- Direct technical initiatives, autonomous robotics workshops, and sensor telemetry testbeds for 40+ engineering students.
- Organize technical bootcamps bridging analytical classroom physics with hands-on hardware laboratory implementation.

A-Level STEM Tutor (Mathematics & Physics) Nov 2022 – Present

Private Mentorship *Remote & In-Person*

- Mentored 20+ students in differential calculus, kinematics, wave mechanics, and electromagnetism, with students achieving top A/A* grades.

HONORS & CERTIFICATIONS

- **Dean's Honor List:** Awarded across 5 semesters for academic excellence at NUST CAE.
- **IBM Professional Certification:** Data Analysis with Python (NumPy, SciPy, Pandas, Scikit-Learn).